

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1250

Yards Creek 138 kV
July 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1250. The Interconnection Customer proposes a 50.4 MW battery energy storage facility interconnecting at the Yards Creek 138 kV substation in the New Jersey area. The request was received on July 14, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 473 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$102,951,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Latigo Substation – Shawnee 138 kV Line	Clear Springs 500/345kV XFMR	138.0	251	19.5
Yards Creek – Latigo 138 kV Line	Clear Springs 500/345 kV Transformer	137.7	244	19.5
Shawnee – Warrick 138 kV Line	Clear Springs 500-345 kV Autotransformer	132.4	262	19.5
Fiberboard – Warrick 138 kV	Clear Springs 500 kV/345 kV Autotransformer	129.5	282	19.5
West Gallatin 345 kV/138 kV Autotransformer	West Gallatin – Felida 345 kV	118.0	375	5.9
Bar Beach 345-138 kV XFMR	Clear Springs 500-345 kV Autotransformer	114.3	365	14.6
Clear Springs 345/138 kV XFMR	Walnut – West Gallatin 345 kV	113.8	439	6.4
Railyard – Central Campus 138 kV	Hills Chapel – Burns 345 kV	113.8	275	6.8
Lakin Municipal – South Hampton 138 kV Line	Wallenpaupack 500-345 kV Autotransformer	113.7	426	6.6
Yards Crk 138 kV/1 kV Auto	Clea Springs 500/345 kV TX	113.3	396	68.3
Lakin Municipal Substation 345 kV/138 kV Transformer Bank	Wallenpaupack 500-345 kV TX	109.3	486	5.0
Central Campus – Fiberboard 138 kV	Burns – Hills Chapel 345 kV	108.7	275	6.8
Hostos 5011 – E Bridgewater 138 kV	Clear Springs 500/345 kV Auto	104.1	261	5.9
East Bridgewater – E 75Th St Substation 138 kV	Fiberboard Substation – Warrick 138 kV Line	103.8	228	8.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gmebird – Mcminn 138 kV	Clear Springs 500/345kV XFMR	103.3	237	5.9
Railyard – South Hampton 138 kV	Wallenpaupack 500/345 kV TX	102.6	466	6.6
Hostos 5011 – Mcminn 138 kV	Clear Springs 500 kV/345 kV TX	102.4	257	5.9
Bar Beach – East 75Th Street 138 kV L	Fiberboard Substation – Warrick 138 kV	100.4	255	8.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Latigo Substation – Shawnee 138 kV Line	Rebuild line with higher-capacity conductor	\$3,744,000	100% (\$3,744,000)
Yards Creek – Latigo 138 kV	Rebuild line with higher-capacity conductor	\$3,309,000	100% (\$3,309,000)
Shawnee – Warrick 138 kV Line	Rebuild line with higher-capacity conductor	\$4,020,000	100% (\$4,020,000)
Warrick – Fiberboard 138 kV Line	Rebuild line with higher-capacity conductor	\$5,889,000	100% (\$5,889,000)
West Gallatin 345 kV/138 kV Autotransformer	Replace transformer with higher-rated unit	\$17,745,000	25% (\$4,379,000)
Bar Beach 345/138kV XFMR	Replace transformer with higher-rated unit	\$14,979,000	100% (\$14,979,000)
Clear Springs 345/138 kV XFMR	Replace transformer with higher-rated unit	\$25,164,000	26% (\$6,613,000)
Railyard – Central Campus 138 kV	Rebuild line with higher-capacity conductor	\$11,988,000	100% (\$11,988,000)
Lakin Municipal – South Hampton 138 kV	Rebuild line with higher-capacity conductor	\$18,384,000	100% (\$18,384,000)
Yards Crk 138 kV/1 kV Auto	Replace transformer with higher-rated unit	\$3,135,000	65% (\$2,035,000)
Lakin Municipal 345-138 kV XFMR	Replace transformer with higher-rated unit	\$10,158,000	100% (\$10,158,000)
Central Campus – Fiberboard 138 kV	Reconductor line	\$8,270,000	100% (\$8,270,000)
Hostos 5011 – E Bridgewater 138 kV	Reconductor line	\$7,820,000	4% (\$330,000)
East Bridgewater – E 75Th St Substation 138 kV	Reconductor line	\$5,201,000	100% (\$5,201,000)
Gamebird – Mcminn 138 kV line	Reconductor line	\$1,346,000	4% (\$57,000)
Railyard – South Hampton 138 kV	Reconductor line	\$2,196,000	100% (\$2,196,000)
Hostos 5011 – Mcminn 138 kV	Reconductor line	\$1,025,000	4% (\$43,000)
Bar Beach – East 75Th Street 138 kV L	Reconductor line	\$1,356,000	100% (\$1,356,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$102,951,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.